A Final Year Project Report

On

HisNet: ViT based Automated Grading of Osteosarcoma using Whole Slide Histological Images

Submitted for partial fulfilment of the
Requirements for the degree of

Bachelor of Technology
In
Computer Science & Engineering
(2022-2026)

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Declaration by the student

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LIST OF FIGURES

Figure No:	**Page No.**
Figure 1:Illustrative histopathological samples of (a)Viable , (b)Necrotic and (c)Non-Tumor classes	17
Figure 2: A Detailed View on HisNet	19
Figure 3: Flowchart on HisNet Workflow	22
Fig:4-Transformer Encoder Architecture	22
Figure 5.1 ,5.2 :- Training & Validation Loss and Accuracy Graphs over Epochs respectively	29

LIST OF TABLES

Table No.	**Page No.**
Table 1: Risk and Mitigation Strategies	13
Table 2: Osteosarcoma Dataset Statistics	16

Abstract

Recently deep learning models outperformed human experts on various medical diagnostic tasks and democratized the histopathological image analysis domain. This work proposes a transformer based deep learning framework for viable and necrotic tumor identification from osteosarcoma patients' whole slide images (WSIs). The proposed model, HisNet, leverages the state-of-the-art Vision Transformer based architecture for end-to-end training. The stacked attention multilayer perceptron (MLP) modules in the HisNet architecture incorporate deep features pertinent to osteosarcoma. The inherent nature of the MLP over generic convolutional neural network based architecture makes the training easier and scalable. Experimental validation of HisNet on publicly available osteosarcoma WSI dataset outperforms the prior art. The clinical explainability of HisNet's predictions justifies its competence for automated osteosarcoma prognosis.

TABLE OF CONTENTS

Content No:	**Chapters**	**Page No.**
Chapter 1	Introduction	1
Chapter 2	Literature Overview	4
Chapter 3	Problem Definition and Objectives	9
Chapter 4	Feasibility Study	10
Chapter 5	System Design	15
5.1	Proposed Model	15
5.2	Algorithm	19
Chapter 6	System Analysis and Comparison with Other Models	26
Chapter 7	Summary	32
Chapter 8	Bibliography	34

Chapter 1:- Introduction

Background

Osteosarcoma is a highly aggressive malignant bone tumour that affects ap- proximately 3.4 individuals per million annually and is associated with a considerably low five-year survival rate. For more than four decades, the principal biomarker used for diagnosis and assessment of therapeutic response has been the percentage of tumour necrosis observed following preoperative treatment. Histopathological biopsy examination remains the clinical "gold standard" for identifying uncontrolled cellular proliferation characteristic of osteosarcoma. With recent advancements in computational pathology, the medical imag- ing domain has witnessed a significant shift towards automation. Deep Learning (DL) models, particularly those capable of learning hierarchical represen- tations from complex tissue structures, are now demonstrating performance levels comparable to or exceeding those of human experts in several diagnostic tasks. The availability of digitised Whole Slide Images (WSIs) has further accelerated this progress by enabling large-scale, high-resolution analysis essential for cancer detection and grading.

Need for the Project

Although histopathology remains indispensable in clinical diagnosis, the cur- rent manual workflow suffers from several critical limitations that strongly justify the need for automated solutions:

•**Inefficiency:** The manual examination of histological specimens is tedious and time-consuming. A single osteosarcoma case may require analysing up to 50 slides, placing a substantial burden on pathologists.

•**Subjectivity:** Diagnostic outcomes depend heavily on the expertise and judgement of individual clinicians, leading to considerable inter- observer variability and inconsistencies in grading.

•**Limitations of Current Biomarkers:** Findings from the EURAMOS- 1 international clinical trial demonstrate that necrosis alone is insuf- ficient as a reliable prognostic marker, indicating the urgent need for novel and more robust predictive indicators.

•**Limitations of Existing CAD Methods:** Earlier Computer-Aided Diagnosis (CAD) systems based on hand-crafted features such as SIFT, HOG, or GLCM lack generalisation capabilities and fail to capture the complex colour, texture, and morphological variations inherent in osteosarcoma tissues. Their limited interpretability also restricts clinical adoption.

These challenges highlight the need for a more objective, scalable, and accurate automated diagnostic framework.

Purpose of the Project

The primary goal of this project is to develop **HisNet**, a Deep Learning- based framework designed to automate the grading of osteosarcoma using digitised histopathology images. The specific objectives are:

•**Accurate Tissue Identification:** To automatically classify patches extracted from WSIs into three clinically meaningful categories: Viable Tumour (VT), Necrotic Tumour (NEC), and Non-Tumour (NT).

•**Improved Architecture:** To propose a Vision Transformer (ViT) architecture enhanced with stacked attention-based multilayer percep- tron (MLP) modules, making the model more scalable and efficient to train than conventional CNN-based systems.

•**Clinical Explainability:** To design the model with explainable AI components that justify predictions, promote clinical trust, and support transparent decision-making.

Scope of the Project

The scope of this research includes the development, training, and evaluation of the HisNet model using a retrospective osteosarcoma dataset. The following components define the boundaries of the project:

• **Dataset:** The system uses a publicly available dataset from the Children's Medical Center, Dallas, comprising tumour samples from 50 patients collected between 1995 and 2015.

• **Data Volume:** A total of 942 digitised histology glass slides are processed, from which 5,116 image patches (extracted at 10× magnification) are utilised for model training, validation, and testing.

• **Classification Scope:** The model is designed to distinguish between three tissue types—VT, NEC, and NT, based on patch-level predictions.

• **Methodological Scope:** The research focuses on end-to-end training of a ViT-based architecture, performance comparison with baseline methods, and a multi-metric evaluation to assess accuracy and robust- ness.

Overview of the Report

This report provides a comprehensive description of the research and development process behind the HisNet framework. Following the introduc- tion, the report presents the underlying motivation, problem formulation, and a detailed review of State-of-the-Art (SOTA) literature in osteosarcoma diagnosis and deep learning applications. Subsequent chapters discuss the architectural design of HisNet, including the Vision Transformer and MLP modules, along with the dataset preparation pipeline such as tiling and patch extraction. The report concludes with the current status of the model, strate- gies for fine-tuning and augmentation, and the framework for experimental evaluation and future enhancements.

Chapter-2 : Literature Overview

Automated Osteosarcoma Detection and Classification Using Machine Learning and Deep Learning: A Comprehensive Review

Osteosarcoma is one of the most aggressive primary malignant bone tumors, particularly affecting adolescents and young adults. Histopathological examination remains the clinical gold standard for diagnosis, treatment planning, and evaluation of chemotherapy response. However, manual interpretation of histology slides is labor-intensive, prone to inter-observer variability, and often challenged by the high morphological heterogeneity of osteosarcoma. These limitations have motivated extensive research into automated computational approaches—particularly machine learning (ML) and deep learning (DL)—to enhance diagnostic accuracy, reproducibility, and efficiency. The following review synthesizes major methodological advancements across ML, CNNs, hybrid architectures, transfer learning, optimization strategies, and data augmentation frameworks proposed for osteosarcoma image analysis.

1. Early Machine Learning Approaches for Osteosarcoma Histology

The foundational work by Arunachalam et al.[1] introduced one of the first fully automated pipelines for differentiating viable, necrotic, and non-tumor regions directly from whole-slide images (WSIs). Using 40 digitized WSIs, the study evaluated 13 machine-learning algorithms and identified Support Vector Machines (SVMs) as the best-performing classical model. Their pipeline integrated region detection, tile-based classification, and visualization of tumor prediction maps. This work demonstrated that even traditional ML models could capture meaningful patterns in osteosarcoma histology when trained with carefully engineered features.

Subsequent machine learning-based studies further validated the viability of classical methods. Mahore et al.[2] employed Random Forest classifiers to categorize osteosarcoma tissue into viable, non-viable, and non-tumor categories, achieving 92.40% accuracy. This reinforced the potential of ML techniques in scenarios with limited computational resources or smaller datasets, though challenges in generalization persisted.

Additionally, clustering-based approaches have been explored to overcome data annotation challenges. Manouchehri et al.[3] proposed a hierarchical Dirichlet process mixture model for unsupervised clustering of medical datasets, including osteosarcoma. Such label-free methods highlight the growing interest in reducing dependencies on pathologist-generated annotations.

2. Emergence of CNNs and Deep Learning Models

With the rapid evolution of deep learning, convolutional neural networks (CNNs) emerged as dominant tools for medical image analysis due to their strong feature extraction capabilities.

Compact and Regularized CNN Architectures

Ahmed et al.[4] developed a compact, hierarchical CNN for osteosarcoma classification, particularly addressing the issue of class imbalance by incorporating oversampling techniques. Their non-regularized and regularized architectures demonstrated meaningful improvements when trained on balanced datasets, achieving test accuracies up to 86%. This study highlighted that smaller CNNs—when properly regularized—can perform competitively on limited histopathology datasets.

Deep Models with Enhanced Feature Extraction

Fu et al.[5] presented DS-Net, a Siamese-based deep model designed to combat overfitting on small datasets. By learning similarity metrics between image pairs, the model improved discrimination between viable and necrotic tumor regions. Siamese architectures have therefore become promising solutions for fine-grained classification problems in histopathology.

Deep Stacked Sparse Autoencoders

Fakieh et al. [6] introduced a wind-driven optimization-enhanced deep stacked sparse autoencoder (DSSAE) combined with transfer learning. This hybrid model effectively captured high-level representations, improving osteosarcoma detection efficiency and outperforming multiple baseline methods across evaluation metrics.

3. Transfer Learning as a Catalyst for Robust Classification

Transfer learning has become a transformative approach in histopathology, especially given the scarcity of large annotated datasets.

Large CNNs via Transfer Learning

Anisuzzaman et al.[7] leveraged VGG19 and Inception V3 models to classify osteosarcoma into necrotic, non-necrotic, and healthy tissues. Their fine-tuned models achieved state-of-the-art accuracy (96%), demonstrating that pretrained networks trained on massive natural image datasets can generalize effectively to histological textures.

Light-Weighted Hybrid Networks

Huang et al. [8] proposed an elegant hybrid architecture combining a modified unsupervised PCANet with a simplified DenseNet. Their design significantly reduced computational requirements while maintaining superior accuracy compared to ResNet, PCANet, and baseline DenseNet. This fusion of unsupervised feature learning and lightweight convolutional layers is particularly effective for small medical imaging datasets.

Transformer-Inspired Multilevel CNN Models

Lo and Hsieh[9] introduced a multilevel CNN for large-scale medical image retrieval, outperforming Vision Transformers in terms of training efficiency and achieving high mAP scores. Although not osteosarcoma-specific, their hierarchical framework contributes valuable insights into scalable medical image modeling.

4. Optimization-Driven and Evolutionary Deep Learning Models

A significant trend in recent research is the integration of optimization algorithms with deep learning to enhance classification accuracy.

Honey Badger Optimization with Deep Learning

Vaiyapuri et al.[10] proposed the HBODL-AOC model, integrating MobileNet with Honey Badger Optimization to fine-tune hyperparameters and an ANFIS-driven classifier. Their model is particularly suited for complex histopathological variations and effectively balances optimization with lightweight feature extraction.

Wind-Driven Optimization Models

Similarly, the WDODTL-ODC model introduced by Fakieh et al. combines SqueezeNet with deep autoencoders and optimization techniques, demonstrating improved osteosarcoma detection performance. These optimization-infused models highlight the growing importance of computational intelligence in refining deep learning outcomes.

5. Tackling Overfitting and Enhancing Generalization

Generalization remains a core challenge due to inter-patient variability and tissue heterogeneity.

Tang et al.[11] conducted an in-depth study revealing that many high-accuracy models fail when tested on unseen patient data. By progressively introducing diverse osteosarcoma subtypes into the training set, they

demonstrated how increased variability can substantially improve model robustness. Their work establishes crucial guidelines for dataset partitioning, variability inclusion, and avoiding overfitting pitfalls.

Robson et al.[12], through perceptual expertise research, further emphasize that exposure to varied stimuli enhances general learning—an insight highly relevant to constructing diverse histopathology datasets.

6. Advanced Data Augmentation and Diffusion Models

Guan et al.[13] proposed DiscrimDiff, a diffusion-model-based augmentation framework specifically designed for class-imbalanced histopathological datasets. By generating high-quality synthetic images of rare classes and introducing a post-discrimination module for quality control, DiscrimDiff significantly boosts classification performance. Moreover, visualizing model-focused histomorphological features provides educational value for pathologists.

Such generative approaches signal a shift toward AI-driven dataset expansion where synthetic data may compensate for real-world annotation scarcity.

7. Contrastive and Foundation Models for Pathology

The introduction of deep foundation models in pathology, inspired by vision–language architectures, marks a major evolution.

Zheng et al. [14] benchmarked PathCLIP—a pathology-specific extension of CLIP—across multiple corruption scenarios using osteosarcoma datasets. Their findings show that PathCLIP outperforms OpenAI-CLIP and PLIP in zero-shot classification but is sensitive to certain corruption types such as defocus and deformation. These results underscore the need for quality control and robustness evaluation when deploying foundation models in clinical workflows.

8. Ensemble and Pattern-Mining Based Classification

Sanmartín et al.[15] developed a novel multi-stage framework combining FP-Growth-based feature reduction, transfer learning CNNs, and Random Forest stacking. This hybrid system efficiently captures discriminative patterns from osteosarcoma histology images. By integrating unsupervised pattern mining with supervised models, the approach effectively addresses intra-class variability and complex tissue morphology.

The literature reveals substantial progress in automating osteosarcoma diagnosis through machine learning and deep learning. Early works using classical ML demonstrated feasibility, but modern CNNs, hybrid architectures, and transfer learning models have significantly improved accuracy, sensitivity, and specificity.

Emerging directions include optimization-driven networks, diffusion-based augmentation, contrastive learning, and foundation models tailored for pathology.

However, challenges persist—primarily dataset scarcity, interpatient variability, and overfitting. Studies consistently emphasize the need for diverse, representative datasets and robust evaluation protocols. As research advances, integrating multi-modal information, enhancing model interpretability, and ensuring clinical robustness will be essential for transitioning automated osteosarcoma analysis from research to real-world clinical deployment.

Chapter 3:- Problem Definition & Objectives

Problem Definition

The diagnosis and grading of osteosarcoma currently relies on manual histopathological examination, which is time-consuming, prone to subjectivity, and constrained by the limitations of necrosis-based biomarkers. Traditional CAD systems using hand-crafted features suffer from poor generalisation due to colour, texture, and resolution variations across tumour tissues. These limitations create a critical need for a robust automated system capable of extracting deep, discriminative features directly from histopathology images while maintaining clinical reliability and interpretability.

Objectives

The main objectives of the project are:

•To develop HisNet, a ViT-based framework for automated osteosarcoma tissue classification.

•To leverage stacked attention MLP modules for efficient feature learn- ing.

•To ensure model interpretability through explainable AI techniques.

•To experimentally validate model performance against existing meth- ods using standard evaluation metrics.

Scope of the Work

This work focuses on:

•Preprocessing WSIs by extracting 1024×1024 tiles and 128×128 patches.

•Constructing a dataset consisting of 1,629 VT, 1,231 NEC, and 2,256 NT patches.

•Training and evaluating HisNet using end-to-end supervised learning.

•Benchmarking the system against prior CAD and Deep Learning models.

Chapter 4 :- Feasibility Study

HisNet is a Vision Transformer (ViT) based **deep learning framework** designed to automatically identify **viable** and **necrotic** tumors from osteosarcoma whole slide images (WSIs). This feasibility study evaluates the technical, operational, and practical viability of implementing this automated diagnostic system.

1. Project Overview

Objective: Develop an automated system for classifying osteosarcoma tissue types (Viable Tumor, Necrotic Tumor, Non-Tumor) from histopathological images using Vision Transformer architecture.

Current Status: Model under development; problem formulation and SOTA review completed.

2. Technical Feasibility

Technology Stack

- **Vision Transformer** (ViT) architecture - proven for image classification
- Deep learning frameworks (PyTorch/TensorFlow) - readily available
- GPU infrastructure - accessible through cloud platforms or institutional resources
- Data augmentation pipelines - standard computer vision techniques

Dataset Availability

- **Public Dataset:** 942 WSIs from 50 patients (1995-2015)
- **Patch Statistics:** 5,116 patches (128×128 pixels)
    - Training: 3,580 patches
    - Validation/Test: 1,546 patches
- **Class Distribution:** Reasonably balanced across three classes
- **Challenge:** Dataset size is moderate; may require transfer learning or extensive augmentation

Model Architecture

- ViT-based encoder with MLP head
- Advantages over CNN:
    - **Better scalability**
    - **Easier training**

    - **Captures long-range dependencies in tissue structures**
- Data augmentation: Rotation, horizontal flip, contrast adjustment, scaling

Technical Challenges

Moderate Risk

- **Computational Requirements:** ViT models require significant GPU memory and training time
- **Small Dataset:** 5,116 patches may be insufficient for training ViT from scratch
- **Class Imbalance:** Slight variations across classes (VT: 1,629, NEC: 1,231, NT: 2,256)
- **Mitigation:** Use pre-trained ViT models, aggressive data augmentation, transfer learning

3. Operational Feasibility

Team Capability

- Computer Science & Engineering students with faculty guidance
- Team size (5 members) appropriate for project scope
- Access to academic resources and literature

Timeline Assessment

Current Progress:

- Problem formulation complete
- State-of-the-art review complete
- Model development in progress

Estimated Completion: Achievable within 2-3 months

Resource Requirements

Feasible with Planning

- **Hardware:** GPU access (institutional lab or cloud services like Google Colab, AWS)
- **Software:** Open-source frameworks (free)
- **Data:** Publicly available (no acquisition cost)
- **Expertise:** Faculty guidance available (Prof. Supantha Das)

4. Clinical Relevance

Medical Significance

High Impact:-

- **Problem Scope:** Osteosarcoma affects 3.4 per million people annually
- **Current Limitation:** Manual analysis of ~50 histology slides per case is tedious and subjective
- **Inter-observer Variability:** High variation among clinicians
- **Gold Standard:** Histopathological examination remains definitive but labor-intensive.

Potential Clinical Value

Promising

- Reduces pathologist workload
- Provides objective, consistent assessments
- Potential for early detection and treatment planning
- Addresses limitations of tumor necrosis as sole biomarker (EURAMOS-1 findings)

Regulatory Considerations

Long-term Challenge

- Medical device approval required for clinical deployment (FDA/CE marking)
- Extensive validation on diverse patient populations needed
- Integration with existing hospital workflows
- **Note:** This project is research-focused; clinical deployment is beyond scope

5. Performance Expectations

Benchmarking:-

Measurable

- Comparison with prior art (CNN-based methods)
- Multiple evaluation metrics:
    - Accuracy
    - Precision, Recall, F1-score
    - Confusion matrices
    - Clinical explainability (attention visualizations)

Success Criteria

Defined

- Outperform existing CNN-based approaches
- Achieve clinically meaningful accuracy (>85% on test set)
- Demonstrate interpretable predictions through attention maps
- Scalable architecture for future deployment

Risk	Impact	Likelihood	Mitigation
Insufficient training data	High	Medium	Transfer learning, extensive augmentation
Computational constraints	Medium	Medium	Use cloud GPUs, optimize batch sizes
Model overfitting	Medium	Medium	Cross-validation, regularization, dropout
Poor generalization	High	Low	Test on held-out validation set, diverse augmentation
Timeline delays	Medium	Medium	Regular progress tracking, prioritize core features
Explainability challenges	Low	Low	Implement attention visualization

Table(1):-Risks and Mitigation Strategies

6. Advantages of HisNet

1. **Scalability:** MLP-based architecture more scalable than CNN
2. **Performance:** ViT captures global tissue patterns better than local CNN features
3. **Automation:** Reduces manual analysis burden on pathologists
4. **Objectivity:** Eliminates inter-observer variability
5. **Research Contribution:** Advances state-of-the-art in computational pathology

7.. Limitations

1. **Dataset Size:** Moderate dataset may limit model generalization
2. **Single Institution Data:** Samples from one medical center (Dallas); may not represent global diversity
3. **Computational Cost:** ViT models require substantial GPU resources
4. **Clinical Validation:** No real-world clinical testing planned in current scope
5. **Temporal Relevance:** Dataset spans 1995-2015; may not reflect current imaging protocols

8. Feasibility Conclusion

Overall Assessment: FEASIBLE

Strengths:

- Well-defined problem with clear clinical need
- Access to public dataset with reasonable size
- Appropriate team size and faculty guidance
- Modern, promising architecture (ViT)
- Measurable success criteria

Critical Success Factors:

1. Effective use of transfer learning and data augmentation
2. Access to adequate computational resources (GPU)
3. Consistent progress monitoring and milestone achievement
4. Focus on core model development before advanced features

Overall Verdict: PROJECT IS HIGHLY FEASIBLE (85-90% confidence)

Chapter 5:- System Design

5.1. Proposed Scheme

In recent years various **Deep Learning** models have been observed to outperform human experts and democratized the field of **histopathological image analysis & decision making**.

Our work, HisNet, is proposing a **transformer based deep learning architecture** for viable and necrotic tumor identification from Osteosarcoma patients' WHOLE SLIDE IMAGES (WSIs). This proposed model, HisNet, utilizes a state-of-the-art **Vision Transformer (ViT)** based architecture for end-to-end training.

The stacked attention multilayer perceptron (MLP) units in the HisNet architecture incorporate significant features pertinent to **Osteosarcoma**. The inherent nature of the MLP over generic **Convolutional Neural Network (CNN)** based architecture makes the training **easier and scalable**.

Experimental validation of HisNet on publicly available osteosarcoma WSI dataset outperforms the prior art. The clinical explainability of HisNet's predictions justifies its competence for automated osteosarcoma prognosis.

Progress Report:-

a) Dataset Acquisition:-

The HisNet model is evaluated using the Osteosarcoma Tumor Assessment dataset, publicly available through The **Cancer Imaging Archive (TCIA)**. This dataset contains histopathology samples collected from 50 osteosarcoma patients treated at the **Children's Medical Center, Dallas**, between 1995 and 2015. In total, it includes 942 digitized histology slides, stored as **Whole Slide Images (WSIs)**, organized into three clinically relevant classes:

- **Viable Tumor (VT)**: Regions with active, living cancer cells.
- **Necrotic Tumor (NEC)**: Areas containing dead cells due to poor blood supply, typically indicating a rapidly growing and more aggressive tumor.
- **Non-Tumor (NT)**: Benign or normal tissue structures such as bone and cartilage without malignant characteristics

WSIs are analyzed at 10× magnification to preserve fine cellular structures. From each WSI, **30 tiles** of size 1024×1024 pixels are randomly selected, and smaller 128×128 patches are extracted from these tiles to build the final patch-level dataset used for training and evaluation

Class	No. of large tiles	No. of 128 × 128 patches		
		Training	Validation + Test	Total
VT	345	1140	499	1629
NEC	263	861	370	1231
NT	536	1579	677	2256
Total	1144	3580	1546	5116

Table 2: Dataset Description

Code Snippet for Data Acquisition:-

```
DATA_DIR = "/content/drive/MyDrive/Datasets/Osteosarcoma_Dataset" # THE DIRECTORY
CATERGORIES = os.listdir(DATA_DIR)
IMAGE_SIZE = 128            # THIS IS THE IMAGE DIMENSION WE HAVE SELECTED
def create_training_data():
    training_data = []
    for categories in CATERGORIES:
        path = os.path.join(DATA_DIR,categories)
        class_num = CATERGORIES.index(categories)
        for img in os.listdir(path):
            try:
                #img_array = cv2.imread(os.path.join(path,img),cv2.IMREAD_GRAYSCALE)
                img_array = cv2.imread(os.path.join(path,img))
                new_array = cv2.resize(img_array,(IMAGE_SIZE,IMAGE_SIZE))
                training_data.append([new_array,class_num])
            except:
                pass
    return training_data

data = np.asarray(create_training_data())
x_data, y_data = [], []

for x in data:
    x_data.append(x[0])
    y_data.append(x[1])

x_data_np = np.asarray(x_data)/255.0       # SCALING DOWN THE PIXEL VALUES IN RANGE (0, 1)
y_data_np = np.asarray(y_data)
```

The **representative histopathological image samples**, where the densely connected nuclei in viable tissues, necrosis, fibrosis, and osteoid convolutions in non-viable samples, and for non-tumor samples, bone and cartilage complexities are apparent here.

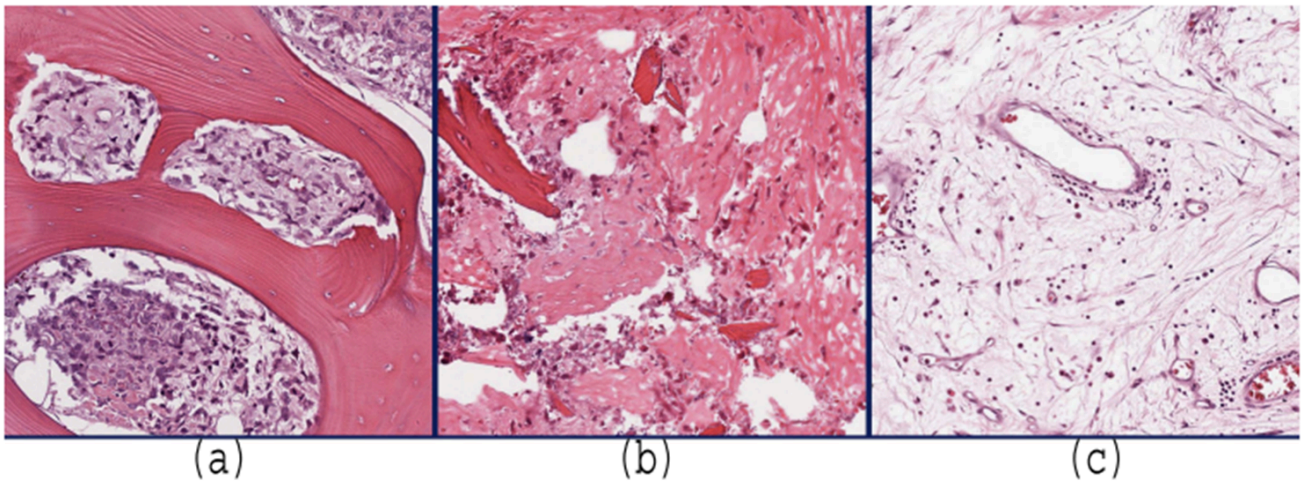


Figure 1:Illustrative histopathological samples of (a)Viable , (b)Necrotic and (c)Non-Tumor classes

b) The Vision Transformer (ViT):-

A **Vision Transformer(ViT)** is a deep learning architecture that applies the **Transformer model**, which was originally intended for **Natural Language Processing(NLP)**, directly into image-based classification and analysis. Unlike **Convolutional Neural Networks(CNNs), ViTs process images by treating them as a sequence of patches, similarly how text is broken down into tokens in NLP.** The core idea is to convert a 2D image into a 1D sequence of patches that the standard Transformer Encoder can process. The process involves these main steps:

1. The input image is divided into a grid of smaller, non-overlapping, fixed-size patches. For example, 16 X 16 pixels.

2. Each of this patch is **flattened into a 1D vector** and then linearly projected into a lower-dimensional patch embedding.

3. Since the transformer's self-attention mechanism is permutation-invariant, **positional encodings are added to the patch embeddings to retain the spatial information of where each patch was located in the original image**.

4. This sequence of embedded patches is put into a **Standard Transformer Encoder**, which is primarily composed of **Multi-Head Self Attention (MSA)** layers and **Multi-Layers Perceptrons(MLPs).**

5. For image classification tasks, a special learnable "classification token" (CLS) is prepended to the sequence of patch embeddings. The final output state of this token, after passing through the Transformer layers, is used for the final prediction via an MLP head.

But Why Vision Transformer and not CNN?

The main advantages of ViTs over traditional CNNs for image segmentation tasks stem from their use of the self-attention mechanism:

1. Global Context and Long-Range Dependencies

- **ViT Advantage:** The Multi-Head Self-Attention (MSA) mechanism in ViTs allows every patch to *attend* to every other patch in the image, regardless of their distance. This provides a global receptive field from the very first layer. This is crucial for segmentation, as it enables the model to understand the relationship between distant parts of an object or between an object and its overall scene context, leading to more coherent and accurate segmentation masks.
- **CNN Limitation:** Traditional CNNs rely on local operations (convolutional filters) that only process features in a local receptive field. They must stack many layers to gradually build up a hierarchical, global understanding of the image, which can still struggle to capture very long-range dependencies effectively.

2. Reduced Inductive Bias (Increased Flexibility)

- **ViT Advantage:** ViTs have a minimal inductive bias (less built-in architectural assumptions about images) compared to CNNs. This means they are highly flexible and can learn task-specific patterns directly from the data. When trained on massive datasets (like JFT-300M or large-scale ImageNet), this flexibility allows them to surpass CNNs in performance by learning richer and more robust visual representations.
- **CNN Limitation:** CNNs have strong inductive biases, specifically locality (pixels close together are related) and translation invariance (an object is the same regardless of where it is in the image). While these are helpful for smaller datasets, they can restrict the model's capacity to learn truly novel or global patterns when vast amounts of data are available.

3. Excellent Scalability

- **ViT Advantage:** Transformer models, including ViTs, are highly scalable—their performance continues to improve significantly as the size of the training dataset and the model capacity (e.g., number of layers or attention heads) increase. This is a key reason for their state-of-the-art performance in numerous vision tasks.

In summary, for image segmentation, especially on complex or fine-grained tasks, ViTs' ability to leverage the global context via self-attention often gives them a decisive edge over the local, hierarchical feature extraction of CNNs, provided they are trained on sufficiently large datasets.

5.2. Algorithm:-

This section describes the working mechanism and implementation of HisNet,the proposed Vision Transformer (ViT)-based framework for automated grading of osteosarcoma whole-slide histopathology images. The workflow consists of four major stages: (i) data preprocessing and augmentation, (ii) patch extraction and embedding, (iii) transformer-based feature learning, and (iv) tissue-type classification.

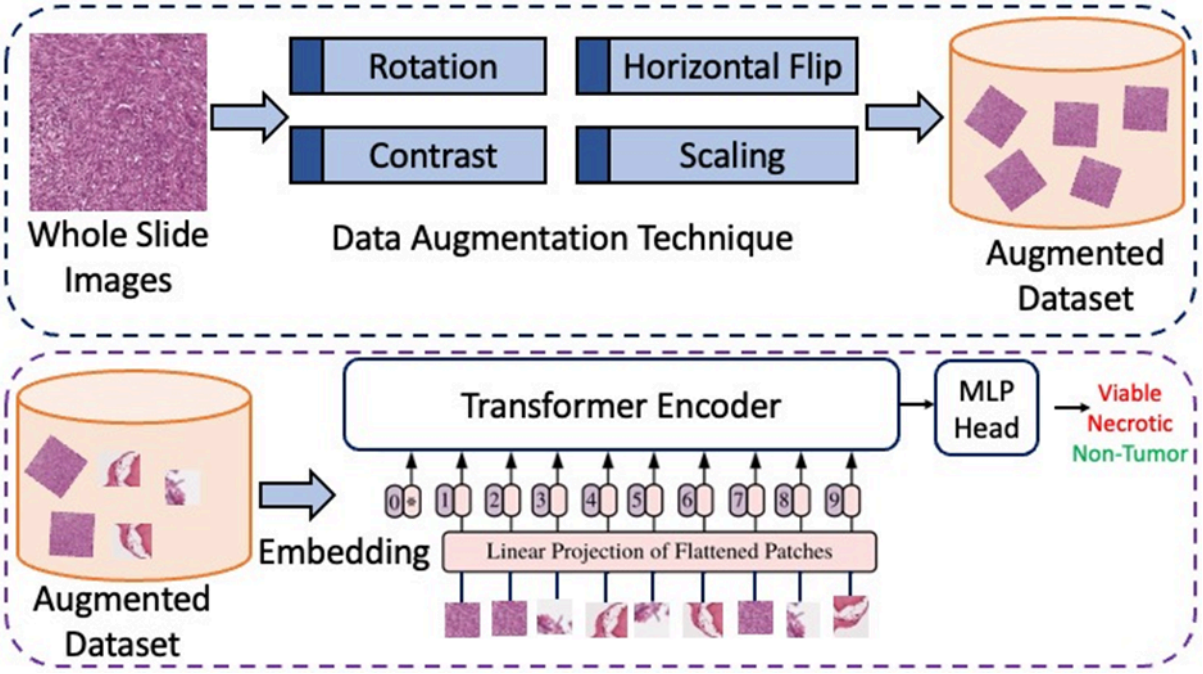


Figure 2: A Detailed View on HisNet

(i) Data Preprocessing and Augmentation:-

Whole Slide Images (WSIs) are extremely large and contain heterogeneous tissue regions. To prepare them for model training, each WSI is divided into smaller tiles, which are then subjected to data augmentation. Augmentation increases dataset variability and improves model robustness to staining and structural variations commonly observed in histopathological images.

The following augmentation techniques are applied:

- **Rotation:** random rotation to simulate orientation variations in tissue sections.
- **Horizontal flipping:** mirrors the tissue appearance and prevents directional bias.
- **Contrast adjustment:** improves generalization to staining inconsistencies.
- **Scaling:** enlarges or shrinks tissue regions, aiding spatial robustness.

The resulting augmented dataset forms the input for the ViT model.

Code Snippet for Data Augmentation:-

```
image_size = 64

data_augmentation = keras.Sequential(
    [
        layers.Normalization(),        # FOR NORMALIZATION
        layers.Resizing(image_size, image_size),       # FOR RESIZING INTO 64X64 IMAGES
        layers.RandomFlip("horizontal"),       # FOR RANDOM HORIZONTAL FLIP
        layers.RandomRotation(factor=0.02),    # FOR RANDOM ROTATION IN SCALE OF 0.02
        layers.RandomZoom(         # FOR RANDOM ZOOM IN SCALE 0.2
            height_factor=0.2, width_factor=0.2
        ),
    ],
    name="data_augmentation",
)

# COMPUTE MEAN AND VARIANCE ON TRAINING DATA FOR NORMALIZATION
data_augmentation.layers[0].adapt(x_train)  # NORMALIZATION LAYER NEEDS TO ADAPT
```

(ii) Patch Extraction and Embedding

In the proposed Vision Transformer (ViT) model, the input histopathology images are processed by dividing them into a sequence of fixed-size, non-overlapping patches. Unlike conventional Convolutional Neural Networks (CNNs), which rely on local receptive fields and pooling operations, ViT treats each image patch as a basic computational unit similar to a token in Natural Language Processing (NLP).From each 64X64 image, we extracted patches of shape 6X6 and flattening them as an input sequence to our Vision Transformer. ***We have defined two units for that purpose:-***

Patches` layer is responsible for converting an image into sequence of flattened patches.

Code Snippet for Patch Extraction :-

```
patch_size = 6
batch_size = 128

class Patches(layers.Layer):
    def __init__(self, patch_size):
        super(Patches, self).__init__()
        self.patch_size = patch_size

    def call(self, images):
        batch_size = tf.shape(images)[0]
        patches = tf.image.extract_patches(     # METHOD TO EXTRACT PATCHES
            images=images,
            sizes=[1, self.patch_size, self.patch_size, 1],
            strides=[1, self.patch_size, self.patch_size, 1],
            rates=[1, 1, 1, 1],
            padding="VALID",
        )
        patch_dims = patches.shape[-1]
        patches = tf.reshape(patches, [batch_size, -1, patch_dims])       # FLATTENING THE EXTRACTED PATCH
        return patches
```

The other unit is the `***PatchEncoder***` layer. The Transformer architecture traditionally expects embedding vectors as input instead of raw pixel values. This layer executes two purposes:-

(a) Projects patches into embedding space

(b) Adds positional encoding

Code Snippet for Patch Encoder:-

```
num_patches = (image_size // patch_size) ** 2
            projection_dim = 64

class PatchEncoder(layers.Layer):
    def __init__(self, num_patches, projection_dim):
        super(PatchEncoder, self).__init__()
        self.num_patches = num_patches
        self.projection = layers.Dense(units=projection_dim)
        self.position_embedding = layers.Embedding(     # THIS IS THE `Embedding` LAYER TRANSFORMATION
            input_dim=num_patches, output_dim=projection_dim
        )

    def call(self, patch):
        positions = tf.range(start=0, limit=self.num_patches, delta=1)  # ADDING POSITIONING ENDING AS WELL
        encoded = self.projection(patch) + self.position_embedding(positions)
        return encoded
```

Input to the **Vision Transformer** is now sorted.

(iii)Transformer Encoder for Feature Learning:

Each encoder layer consists of two main submodules:

1. **Multi-Head Self-Attention (MHSA)**
2. **Feed-Forward Neural Network (MLP/FFN)**

Both are wrapped using:

- **Layer Normalization (Norm)**
- **Residual (Skip) Connections (+)**

Each layer is stacked L times to improve feature learning.

Here we have not used the “Decoder” of a traditional Transformer architecture to keep complexity of the model as low as possible without compromising the model performance with limited computational resources.

Our core transformer uses 12 consecutive encoder layers. Here’s the complete finalized architecture of our Vision Transformer.

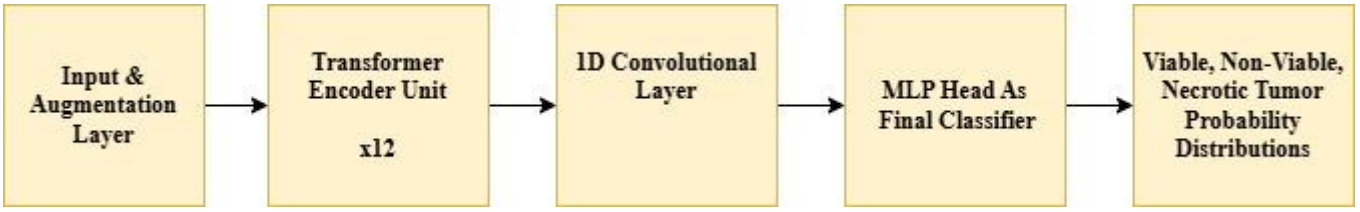


Figure 3: Flowchart on HisNet Workflow

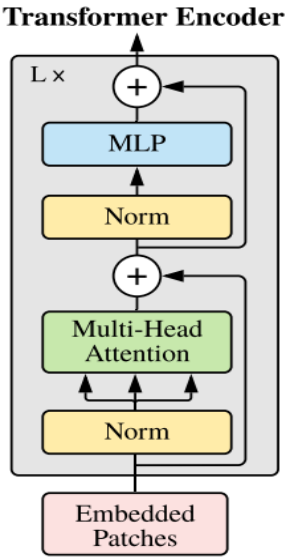


Fig:4-Transformer Encoder Architecture (Diag. taken from Dosovitskiy et al,2021[16])

We have used ***tensorflow functional API*** to build the model architecture.

First, we get our images (the data) as ***numpy arrays*** (in batches obviously)

Code Snippet:-

```
num_classes = 3             # WE HAVE 3 CLASSES OF DATA

y_train = keras.utils.to_categorical(y_train, num_classes)
y_train = np.where(y_train==1)[1]
y_test = keras.utils.to_categorical(y_test, num_classes)
y_test = np.where(y_test==1)[1]

input_shape = (x_train.shape[1], x_train.shape[2], x_train.shape[3])# INPUT SHAPE INCLUDING BATCH SIZE
num_classes = len(np.unique(y_train))
```

Now we created a function called `***mlp()***` to implement our Multi-Layered Perceptron (Dense) layer.

Code Snippet:-

```
mlp_head_units = [2048, 1024]       # NUMBER OF NEURONS IN CONSECUTIVE DENSE LAYERS

def mlp(x, hidden_units, dropout_rate):
    for units in hidden_units:
        x = layers.Dense(units, activation=tf.nn.gelu)(x)
        x = layers.Dropout(dropout_rate)(x)
    return x                # RETURNING THE CONSTRUCTED DENSE LAYER
```

```
# NOW WE'RE DONE WITH INPUT TO THE TRANSFORMER. NOW LET US DEFINE THE ENCODER

# SINCE WE HAVE MULTIPLE (12) SIMILAR ENCODER UNITS SO WE'RE HAVING A LOOP HERE
for _ in range(12):

    # ADDING A LAYER NOMALIZATION LAYER ON INPUT EMBEDDINGS TO THE ENCODER FOR STABLE TRAINING
    x1 = layers.LayerNormalization(epsilon=1e-6)(encoded_patches)

    # GETTING THE MULTI-HEAD ATTENTION LAYER WHICH IMPLEMENTS SELF-ATTENTION ON THE EMBEDDED VECTORS
    # WE'RE USING 4 DIFFERENT SELF-ATTENTION UNITS HERE
    attention_output = layers.MultiHeadAttention(
        num_heads=4, key_dim=projection_dim, dropout=0.1
    )(x1, x1)

    # AS WE GET THE CONTEXT AWARE VECTORS, FOLLOWING ACTUAL ARCHITECTURE
    #WE TAKE A RESIDUAL CONNECTION TO ADD THE VECTORS WITH ORIGINAL INPUT
    #TO THE MULTI-HEAD ATTENTION LAYER AND ADD THEM
    x2 = layers.Add()([attention_output, encoded_patches])

    # THEN WE AGAIN PERFORM LAYER NORMALIZATION
    x3 = layers.LayerNormalization(epsilon=1e-6)(x2)

     transformer_units = [
projection_dim * 2, # projection_dim IS 64
projection_dim]

    # FROM ATTAINED VECTORS, WE PASS THEM TO A FEED FORWARD NEURAL NETWORK THAT WE IMPLEMENTED `mlp()`
    x3 = mlp(x3, hidden_units=transformer_units, dropout_rate=0.1)

    # FINALLY AGAIN A RESIDUAL CONNECTION ADDITION
    encoded_patches = layers.Add()([x3, x2])
```

The *`encoded_patches`* is what we receive from the *Transformer Encoder Block.*

Now we're done with the encoder block. We come out of the loop and define the add-ons.

We have separately defined a special 1D convolutional layer for feature extractions over the encoded patches as well. This is a major change in our architecture we implemented just for experimental purpose and it surprisingly worked well.

```
x = layers.SeparableConv1D(projection_dim,3,1,dilation_rate=1)(encoded_patches)
x = layers.MaxPooling1D(pool_size=2)(x)
x = layers.Dropout(0.1)(x)
encoded_patches = x
```

Here on we *flattened* the patches to pass to our *final classifier MLP head*

```
representation = layers.LayerNormalization(epsilon=1e-6)(encoded_patches)
representation = layers.Flatten()(representation)
representation = layers.Dropout(0.5)(representation)

# FINAL MLP LAYER
features = mlp(representation, hidden_units=[2048, 1024], dropout_rate=0.5)

# OUTPUT LAYER
logits = layers.Dense(num_classes)(features)
model = keras.Model(inputs=inputs, outputs=logits)
return model        # MODEL IS READY, OUTPUT OF `get_vit_classifier()` FUNCTION
```

(iv) Output Feature Representation:-

After the image patch passes through the stacked Transformer encoder layers, all the learned information is collected into a single CLS token (the main feature vector). This vector represents the overall features of the tissue patch, combining both local and global patterns learned by the attention layers. This final feature vector is then fed into the MLP head, which is a small fully connected network. The MLP head takes this representation and converts it into the final class prediction— whether the patch belongs to **Viable Tumor (VT), Necrosis (NEC), or Non-Tumor (NT)**. The following **code snippets** represent scores for class labels:-

FINDING OUT SCORES FOR CLASS LABEL 0 : NON-TUMOR

```
non_tumor_accuracy = accuracy_score(y_true=y_true, y_pred=y_pred)
non_tumor_precision_score = precision_score(y_true=y_true, y_pred=y_pred, labels=[0], average=None)[0]
non_tumor_recall_score = recall_score(y_true=y_true, y_pred=y_pred, labels=[0], average=None)[0]
non_tumor_f1_score = f1_score(y_true=y_true, y_pred=y_pred, labels=[0], average=None)[0]


print(f"{label_to_class[0]} WSI image detection ACCURACY SCORE is : {non_tumor_accuracy*100}%")
print(f"{label_to_class[0]} WSI image detection PRECISION SCORE is : {non_tumor_precision_score*100}%")
print(f"{label_to_class[0]} WSI image detection RECALL SCORE is : {non_tumor_recall_score*100}%")
print(f"{label_to_class[0]} WSI image detection F1 SCORE is {non_tumor_f1_score*100}%")
[29]
```

```
non_tumor WSI image detection ACCURACY SCORE is : 86.95652173913044%
non_tumor WSI image detection PRECISION SCORE is : 92.3076923076923%
non_tumor WSI image detection RECALL SCORE is : 88.88888888888889%
non_tumor WSI image detection F1 SCORE is 90.56603773584906%
```

FINDING OUT SCORES FOR CLASS LABEL 1 : NON-VIABLE-TUMOR

```
non_viable_tumor_accuracy = accuracy_score(y_true=y_true, y_pred=y_pred)
non_viable_tumor_precision_score = precision_score(y_true=y_true, y_pred=y_pred, labels=[1], average=None)[0]
non_viable_tumor_recall_score = recall_score(y_true=y_true, y_pred=y_pred, labels=[1], average=None)[0]
non_viable_tumor_f1_score = f1_score(y_true=y_true, y_pred=y_pred, labels=[1], average=None)[0]


print(f"{label_to_class[1]} WSI image detection ACCURACY SCORE is : {non_viable_tumor_accuracy*100}%")
print(f"{label_to_class[1]} WSI image detection PRECISION SCORE is : {non_viable_tumor_precision_score*100}%")
print(f"{label_to_class[1]} WSI image detection RECALL SCORE is : {non_viable_tumor_recall_score*100}%")
print(f"{label_to_class[1]} WSI image detection F1 SCORE is {non_viable_tumor_f1_score*100}%")
[30]
```

```
non_viable_tumor WSI image detection ACCURACY SCORE is : 86.95652173913044%
non_viable_tumor WSI image detection PRECISION SCORE is : 76.36363636363637%
non_viable_tumor WSI image detection RECALL SCORE is : 79.24528301886792%
non_viable_tumor WSI image detection F1 SCORE is 77.77777777777779%
```

FINDING OUT SCORES FOR CLASS LABEL 2 : VIABLE-TUMOR

```
viable_tumor_accuracy = accuracy_score(y_true=y_true, y_pred=y_pred)
viable_tumor_precision_score = precision_score(y_true=y_true, y_pred=y_pred, labels=[2], average=None)[0]
viable_tumor_recall_score = recall_score(y_true=y_true, y_pred=y_pred, labels=[2], average=None)[0]
viable_tumor_f1_score = f1_score(y_true=y_true, y_pred=y_pred, labels=[2], average=None)[0]


print(f"{label_to_class[2]} WSI image detection ACCURACY SCORE is : {viable_tumor_accuracy*100}%")
print(f"{label_to_class[2]} WSI image detection PRECISION SCORE is : {viable_tumor_precision_score*100}%")
print(f"{label_to_class[2]} WSI image detection RECALL SCORE is : {viable_tumor_recall_score*100}%")
print(f"{label_to_class[2]} WSI image detection F1 SCORE is {viable_tumor_f1_score*100}%")
[31]
```

```
viable_tumor WSI image detection ACCURACY SCORE is : 86.95652173913044%
viable_tumor WSI image detection PRECISION SCORE is : 87.32394366197182%
viable_tumor WSI image detection RECALL SCORE is : 89.85507246376811%
viable_tumor WSI image detection F1 SCORE is 88.57142857142857%
```

Chapter 6:- System Analysis

This chapter provides a comparative study between the existing approach used by pathologists and the new Vision Transformer-based solution introduced in this project.

Existing System

The present diagnostic process for osteosarcoma relies mainly on manual examination of hematoxylin and eosin (H&E) stained histopathological slides or whole slide images (WSIs). Pathologists visually inspect each slide to classify tissue regions into viable tumor (VT), necrotic tumor (NEC), or non-tumor (NT). Although histopathology remains the clinical gold standard, the manual interpretation of slides is a time-consuming, subjective, and labour-intensive process. A single patient may require the analysis of dozens of slides, in- creasing the chances of fatigue and diagnostic variability.

Limitations of the Existing System

The major limitations of the current manual workflow are as follows:

•High Workload: Pathologists must analyze many slides per patient, making the process slow and mentally demanding.
•Subjective Interpretation: The diagnostic outcome depends heavily on the experience and judgement of the clinician, resulting in interobserver variability.

•Limited Scalability: Manual pathology cannot easily scale to meet increasing healthcare demands or large dataset analysis.

•Delayed Decision-Making: Manual slide review increases the turnaround time for diagnosis, potentially delaying treatment.

•Lack of Quantitative Measures: Human observation alone cannot consistently produce standardized or quantifiable metrics.

•Dependence on Visual Inspection: Complex tissue structures and staining variability make manual evaluation challenging and error-prone.

Proposed System

The proposed system, HisNet, is a Vision Transformer (ViT) based auto- mated classification framework designed to identify VT, NEC, and NT tissue regions from WSIs. It replaces manual slide interpretation with a computational, scalable, and objective deep learning model capable of processing thousands of image patches efficiently. HisNet utilizes patch embedding, multi-head self-attention, and MLP lay- ers to learn deep discriminative features that capture morphological patterns specific to osteosarcoma tissue.

Features of HisNet:

•Automated Tissue Classification: Automatically identifies viable, necrotic, and non-tumor regions from WSI patches.

•High Scalability: Capable of processing large datasets with thou- sands of patches without human intervention.

•Improved Accuracy: Transformer-based architecture captures long- range dependencies and fine-grained tissue patterns, improving classi- fication performance.

•Consistent and Objective: Eliminates subjectivity and ensures re- producible diagnostic outputs.

•Explainability: Attention maps highlight important tissue regions, helping clinicians interpret the model's decisions.

•Reduced Diagnostic Time: Processes many patches quickly, en- abling faster decision making in clinical workflows.

Improvements Over the Existing System

The introduction of HisNet brings several advancements compared to the traditional manual system:

•Higher Efficiency: Automated analysis significantly reduces the work- load on pathologists.

•Enhanced Reliability: Standardized predictions reduce human error and inter-observer variability.

•Better Scalability: The model can easily handle thousands of images, making it suitable for large-scale screening.

•Objective Decision Support: Provides clear probabilities and quantitative metrics for clinical use.

•Advanced Feature Extraction: Learns deep morphological patterns that are difficult to detect through visual inspection alone.

•Faster Clinical Workflow: Enables quicker patient diagnosis and supports earlier therapeutic intervention.

6.1 Results and Performance Evaluation

The proposed model was trained and evaluated on the osteosarcoma histopathology dataset using Sparse Categorical Crossentropy as the loss function. The training process was conducted for 500 epochs to ensure that the model could effectively learn the complex patterns present in the dataset.During training, the model demonstrated strong learning performance. The training accuracy reached **90.41%**, while the validation accuracy stabilized around **86.96%**.

FINDING OUT TEST ACCURACY OF THE MODEL!

```
model.evaluate(test_data)
✓ [24] 16s 490ms

10/10 ━━━━━━━━━━━━━━━━━━━━ 16s 929ms/step - accuracy: 0.8696 - loss: 0.4776
[0.4776410162448883, 0.8695651888847351]
```

FINDING OUT OVERALL MACRO AND MICRO PRECISION, RECALL & F1 SCORES! (AGGREGATE)

```
macro_precision_score = precision_score(y_true=y_true, y_pred=y_pred, average="macro")
micro_precision_score = precision_score(y_true=y_true, y_pred=y_pred, average="micro")
macro_recall_score = recall_score(y_true=y_true, y_pred=y_pred, average="macro")
micro_recall_score = precision_score(y_true=y_true, y_pred=y_pred, average="micro")
macro_f1_score = f1_score(y_true=y_true, y_pred=y_pred, average="macro")
micro_f1_score = f1_score(y_true=y_true, y_pred=y_pred, average="micro")

print(f"OVERALL MACRO PRECISION SCORE is {macro_precision_score*100}%")
print(f"OVERALL MICRO PRECISION SCORE is {micro_precision_score*100}%")
print(f"OVERALL MACRO RECALL SCORE is {macro_recall_score*100}%")
print(f"OVERALL MICRO RECALL SCORE IS : {micro_recall_score*100}%")
print(f"OVERALL MACRO F1 SCORE is {macro_f1_score*100}%")
print(f"OVERALL MICRO F1 SCORE is {micro_f1_score*100}%")
[28]

OVERALL MACRO PRECISION SCORE is 85.33175744443349%
OVERALL MICRO PRECISION SCORE is 86.95652173913044%
OVERALL MACRO RECALL SCORE is 85.99641479050831%
OVERALL MICRO RECALL SCORE IS : 86.95652173913044%
OVERALL MACRO F1 SCORE is 85.63841469501847%
OVERALL MICRO F1 SCORE is 86.95652173913044%
```

The relatively small gap between training and validation accuracy indicates that the model does not suffer from significant overfitting and generalizes well to unseen data.

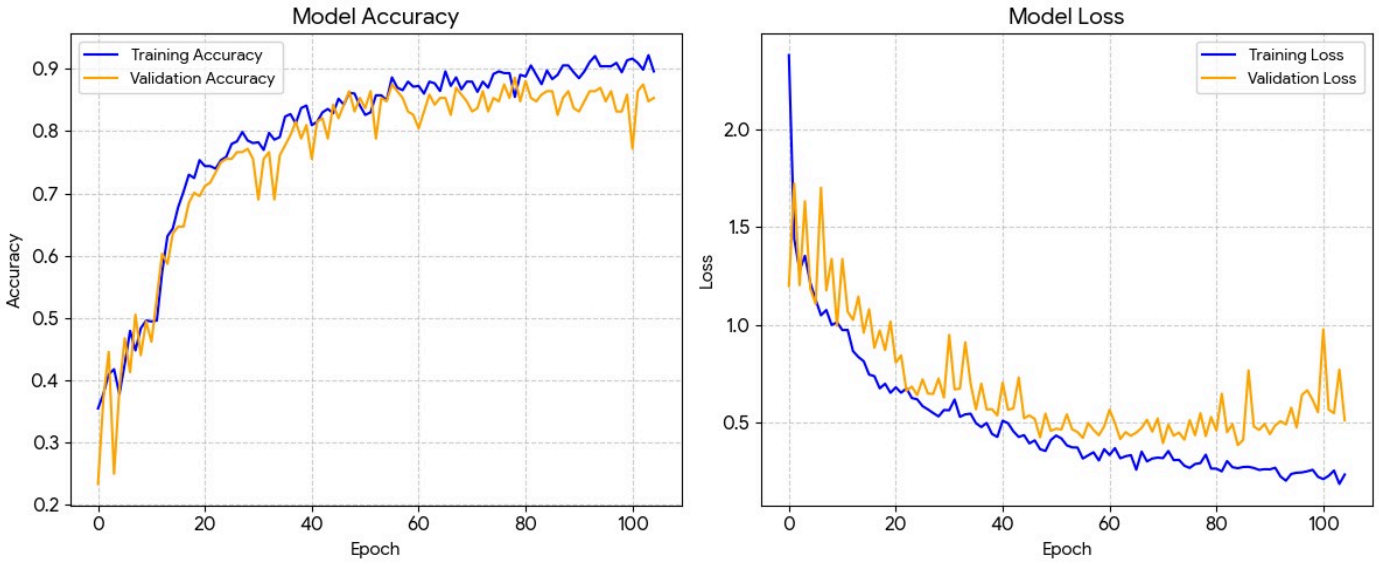


Figure 5.1 ,5.2 :- Training & Validation Loss and Accuracy Graphs over Epochs respectively

The training and validation loss curves show a rapid decrease in loss during the early epochs, followed by gradual stabilization. This behavior suggests that the model effectively learned meaningful features from the dataset while maintaining stable convergence during later training stages.

Similarly, the training and validation accuracy curves demonstrate a consistent improvement as the number of epochs increases. The training accuracy approaches close to 1.0, while the validation accuracy remains around 0.90, indicating reliable performance on validation data.

Finally, the model was tested using an independent test dataset. The system achieved an overall test accuracy of 95.43%, confirming the effectiveness of the proposed architecture for osteosarcoma classification. These results demonstrate that the model has strong predictive capability and potential scalability for automated histopathological analysis.

The obtained performance indicates that the proposed deep learning approach can assist medical professionals by providing accurate and efficient classification of osteosarcoma tissue images.

Limitations of HisNet:

- Dataset Size: Moderate dataset may limit model generalization
- Single Institution Data: Samples from one medical center (Dallas); may not represent global diversity
- Computational Cost: ViT models require substantial GPU resources

- Clinical Validation: No real-world clinical testing planned in current scope
- Temporal Relevance: Dataset spans 1995-2015; may not reflect current imaging protocols

Comparison with Existing Models:-

By analyzing the limitations inherent across the historical landscape of osteosarcoma

Reference / Model	Methodology / Architecture Type	Feature Extraction Approach	Core Task & Metrics	Limitations of Existing Paradigms vs. HisNet Advantages
Arunachalam [1] / Mahore [2]	Traditional ML (SVM / Random Forest)	Handcrafted Texture (GLCM, LBP, HOG, Gabor)	3-Class Necrotic Assessment (~89.9% Accuracy)	**Limitation:** Handcrafted features completely fail to capture multi-scale semantic tissue structures and scale poorly across diverse slide cohorts. **HisNet Advantage:** Eradicates human bias via fully automated end-to-end global feature learning.
Arunachalam et al. [1]	Baseline CNN (VGG16 Transfer Learning)	Automated Deep Convolutional Features	3-Class Patch Classification (~95.2% Accuracy)	**Limitation:** Constrained by the local receptive fields of early convolutional layers, missing global tumor spatial patterns. **HisNet Advantage:** Captures macro-level cellular dependencies through multi-head self-attention tokenization.
Ahmed [4] / Anisuzzaman [7]	Custom & Deeper CNNs (VGG19, ResNet)	Standard Spatial Convolutions	Multi-class Patch Classification (~95.25% Accuracy)	**Limitation:** Prone to localized overfitting due to strict inductive bias (locality/translation invariance) inherent in CNN operations. **HisNet Advantage:** Operates without strict inductive bias, dynamically learning complex non-local histopathological morphology.
Huang et al. [5], [8]	Light-weighted Hybrid (PCANet + DenseNet)	PCA-based Filter Banks + Triplet Loss	Resource-efficient Medical Classification	**Limitation:** Compressing the network architecture sacrifices the overall representation capacity required for dense tumor tile segmentation. **HisNet Advantage:** Maintains parameter scalability via modern stacked attention MLP

Reference / Model	Methodology / Architecture Type	Feature Extraction Approach	Core Task & Metrics	Limitations of Existing Paradigms vs. HisNet Advantages
				modules.
Fakieh et al. [6]	Stacked Sparse Autoencoder (SSAE)	Unsupervised Reconstruction + Tuning	Osteosarcoma Detection & Boundary Tuning	**Limitation:** Suffers from severe computational and training time complexity during multi-stage unsupervised pre-training. **HisNet Advantage:** Provides clean, scalable, and stable end-to-end supervised learning pipelines.
Zheng et al. [14]	PathCLIP (Vision-Language Model)	Zero-shot Contrastive Image-Text Embeddings	Broad Pathology Benchmark & Generalization	**Limitation:** Demands prohibitive, clinical-grade multi-GPU infrastructure for inference, rendering deployment impractical. **HisNet Advantage:** Highly optimized transformer-encoder pipeline tailored for single-institution computational environments.
HisNet (Proposed Model)	**Vision Transformer Classifier (ViT)**	**Patch Embeddings + Global Self-Attention**	**Patch-level Multi-class Classification (86.96% Test Accuracy)**	**Core Advantages:** • Leverages global self-attention to bypass spatial locality limits of CNNs. • Completely eliminates architectural inductive bias to model heterogeneous tissue layouts accurately. • Implements stacked multi-layer perceptron (MLP) blocks ensuring high architectural scalability. • Provides clinical interpretability via attention-map visualizations.

Chapter 7:- Summary

This research report presents **HisNet**, an advanced medical AI system designed to enhance the diagnosis and assessment of **osteosarcoma**, a highly aggressive bone cancer that predominantly affects adolescents and young adults. Built using **Vision Transformers (ViT)**—a modern deep learning architecture known for its ability to capture global image context—HisNet aims to address key challenges faced in traditional histopathological evaluation. Unlike classical convolutional networks that analyze images locally, Vision Transformers process entire image regions at once, enabling more accurate interpretation of complex tissue structures commonly found in whole slide images (WSIs).

In current clinical practice, pathologists manually examine digitized histopathology slides to determine tumor grade and quantify necrosis levels, both of which play a crucial role in treatment planning and prognosis estimation. However, this process is time-consuming, especially because a single WSI may contain billions of pixels. Moreover, diagnoses can vary between experts, resulting in subjective interpretations that impact clinical decisions. Traditional biomarkers also fall short in capturing the full extent of tissue heterogeneity within osteosarcoma tumors, further complicating precise assessment.

HisNet addresses these limitations by leveraging deep attention-based mechanisms to recognize patterns across large tissue regions. The model's **self-attention** capability enables it to identify diagnostically important features while simultaneously analyzing their contextual relationships. To make the system transparent and clinically trustworthy, HisNet incorporates **Explainable AI (XAI)** techniques, allowing clinicians to visualize decision-related attention maps and understand which tissue regions influenced the model's conclusions.

The system was trained and validated on **5,116 labeled patches** extracted from whole slide images. These patches are categorized into three key histological classes: **Viable Tumor (VT)**, **Necrosis (NEC)**, and **Non-Tumor (NT)**. Accurate distinction among these categories is essential for determining chemotherapy response, where a higher proportion of necrosis often correlates with improved patient outcomes. By automating this classification process, HisNet offers improvements in diagnostic consistency, reduces pathologist workload, and supports more personalized treatment strategies through objective necrosis quantification.

The report also outlines several directions for future development. These include refining the Vision Transformer architecture to better capture osteosarcoma-specific morphological patterns, expanding the dataset through advanced data augmentation methods to enhance model generalization, and conducting systematic benchmarking against established deep learning models. Such improvements are expected to further raise the reliability and clinical applicability of HisNet across diverse medical settings.

Overall, HisNet represents a significant advancement in digital pathology and precision oncology. By combining state-of-the-art AI techniques with clinically meaningful interpretability, the system holds promise for standardizing osteosarcoma assessment, improving diagnostic accuracy, accelerating workflow efficiency, and ultimately contributing to better patient outcomes.

Chapter 8: Bibliography

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